

KESAI: Kyutai ELLIS Scalable Autonomous Intelligence

The [ELLIS Institute Tübingen](#) and [kyutai Paris](#) are launching a joint non-profit open science frontier AI research lab co-located in Tübingen and Paris under the leadership of Andreas Geiger, Kashyap Chitta and Bernhard Schölkopf and co-founded by Bernhard Jaeger and Daniel Dauner. This Franco-German flagship collaboration merges kyutai's open science ethos and foundation model expertise with ELLIS's talent attraction and foundational research excellence in 3D computer vision, data-driven simulation, causality and physical AI.

The goal: to pioneer world-leading research in world models and autonomy, and to develop the first fully-open competitive self-driving stack. Towards this goal, the research lab will develop data and compute-efficient autonomous intelligence via latent world models for reasoning and decision-making in complex environments and advance the frontier of physical AI.

Challenge: AI research, especially in self-driving, is dominated by proprietary silos (Waymo, Tesla, Waabi, Wayve, Baidu, Huawei) with closed code, models and data. Current approaches require billion-scale investments and rely on very large data, compute and human resources.

Opportunity: KESAI aims to create a flexible, open, robust, and platform-independent self-driving stack, facilitating manufacturers to integrate state-of-the-art level 5 autonomy. By advancing the frontier of simulation-based physical AI, the developed approach will be several orders of magnitude more efficient in terms of data, compute and human resources.

Impact: The lab will not only publish research but also develop disruptive, fully open-source technology (open data, models, code, documentation and research papers), accelerating innovation in data- and compute-efficient physical AI.

Talent: KESAI attracts top researchers by providing a protected environment focused on scientific breakthroughs, offering open publishing, spin-off opportunities, and competitive, industry-benchmarked salaries without the constraints of corporate or academic institutions.

Technology: KESAI will research data- and compute-efficient methods to develop versatile and robust foundation models for physical AI. In contrast to the dominant data-driven approaches, it will research massively parallel hybrid and latent world models, while leveraging sim-2-real techniques for real-world deployment. The hybrid nature of these models allows for ingesting information from a variety of sources, including real-world data, privileged simulations, and common sense knowledge embedded in large language models. Their latent nature enables data- and compute-efficient closed-loop training of robot foundation policies. This enables zero or few-shot generalization to a large diversity of hardware setups, environments, and rare situations. Self-driving is the epitome of physical AI, covering multi-modal perception, planning in safety-critical situations, and control in highly dynamic environments. KESAI will demonstrate the capabilities of its models by training level 5 self-driving policies using significantly less data and compute than typically required while reaching infraction rates competitive with frontier systems. Based on this foundation, KESAI will extend its effort to other areas of robotics. All code and models will be released under permissive licenses that allow civil commercial use.